

Sankalp Kumar

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PROFESSIONAL SUMMARY

Computer Science undergraduate (2024-28) specializing in full-stack web development and AI-integrated software engineering. Experienced in architecting robust backends, deploying machine learning models, and building scalable web platforms. Eager to transition from building complex independent applications to engineering production-level systems within a collaborative, high-standards team.

EDUCATION

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

Kattankulathur, Tamil Nadu

Aug 2024 – Expected May 2028

- CGPA: 9.49 / 10
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Web Technologies, Operating Systems

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C, C++, HTML5, CSS3

Frontend: React 19, Tailwind CSS, Framer Motion

Backend & Cloud: Node.js, Express.js, FastAPI, MongoDB, SQLite, Firebase

AI & Tooling: TensorFlow, Google Gemini API, Ollama, Git, WebSockets, REST APIs, JWT

PROJECTS

ManagerZero (AI Chief of Staff) | *React, Node.js, Express, Gemini API, Resend*

- Built a full-stack platform to automate engineering management overhead by processing daily standups, extracting meeting action items, and analyzing client feedback sentiment.
- Engineered a backend orchestration layer routing data to 12 specialized Gemini AI agents (e.g., Blocker Detector, Churn Predictor) to generate targeted insights and sprint health scores.
- Secured the REST API using express-rate-limit (1000 req/15 min) and express-validator, integrating the Resend API to dispatch automated email briefings.

Jan Sahayak: AI Civic Issue Reporter | *React.js, FastAPI, TensorFlow, SQLite*

- Engineered a web-based civic action platform utilizing React.js, enabling citizens to report urban hazards and utilizing geospatial algorithms to automatically detect duplicate reports within a 50-meter radius.
- Architected a FastAPI backend integrating a custom-trained TensorFlow CNN to classify user-uploaded images and dynamically route complaints to the correct municipal department.
- Implemented a gamification engine utilizing SQLite to track citizen contributions, calculate Civic Points, and manage real-time community leaderboards.

Browser-Based IDE with AI Assistance | *React, Node.js, WebSockets, Ollama*

- Built a web-based code editor utilizing WebSockets to stream real-time standard output (stdout) and errors during code execution to reduce client-server latency.
- Implemented sandboxed execution environments on the backend to safely run unverified, user-submitted code in isolation.
- Hosted a local Ollama instance on the server to generate code completion suggestions natively, eliminating external API rate limits.

OMNIBRAIN: Multi-Class Heart Risk Predictor | *Machine Learning*

- Trained a classification model using the heart_disease_uci.csv dataset to categorize patient risk across a 5-level severity scale (from Class 0: Healthy to Class 4: Critical).
- Cleaned and normalized raw dataset inputs, isolating heavily weighted physiological features like Maximum Heart Rate (thalach) to improve prediction accuracy.

HACKATHONS & ACHIEVEMENTS

Smart India Hackathon (SIH) Participant

Aug 2025

Heritage Tourism Domain

- Collaborated on a technical proposal to digitize heritage tourism for Sikkim monasteries, focusing on database architecture and user access.

Hack the Knight Participant

Apr 2026

Competitive Hackathon

- Competed in a 48-hour sprint to architect and deploy a minimum viable product (MVP), adapting to new backend frameworks on the fly to meet strict deployment deadlines.

CERTIFICATIONS

- NPTEL – Database Management Systems (2025)
- NPTEL – Programming in Java (2025)